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Presentation Time: 9:00 AM-6:30 PM

GIS AND STATISTICAL ANALYSIS OF RESIDENTS RESPONSE TO VOLCANIC HAZARDS ON NISYROS, SOUTH AEGEAN

JONES, Morgan Bridget¹, MARSELLOS, Antonios¹ and KYRIAKOPOULOS, Konstantinos², (1)Department of Geology, Environment, and Sustainability, Hofstra University, 1000 Fulton Ave, Hempstead, NY 11549, (2)Department of Geology and Geoenvironment, National & Kapodistrian University of Athens, Ilissia, GR 15701, Athens, 15701, Greece, mjones19@pride.hofstra.edu

Nisyros is a volcanic Aegean island situated between Kos and Tilos. Nisyros has a diameter of about 8km and an area of 41.6km². The island contains a 3km wide caldera where the volcano is currently active with a previous past of four historical phreatic eruptions. The island hosts a population of about 1,008 persons spread between the main town of Mandraki, Paloi, Nikia, and Emporeios. However, the island is heavily trafficked by tourists during the summer months.

To evade future damage and possible loss of life, it is necessary to determine subsequent areas of high risk prone to volcanic events and alert residents. A statistical and GIS analysis of surveys were conducted to assess the preparedness and interest of the local community for an online warning system. Research found that there lacked an official emergency evacuation route and other safety measures, should a volcanic eruption occur. Twenty residents were randomly selected to answer a series of questions to see if volcanic monitoring information would be useful. Using a statistical software (SPSS), a Chi-square test was obtained to test the dependency of any categorical variable pairs while Cramer's V and Phi test were used to test the relationship strength between the questions asked. Additionally, a Kriging Interpolation prediction surface map was created from the surveys to see if there were any spatial autocorrelations. The p-values from the tests showed that there was a dependency where most residents felt unprepared for any hazards and were interested in an app that could monitor and assist in any volcanic activity on the island.

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